

Release numerics update

The ionization closure, the domain, and the new face solver

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- 1 Which ionization closure, and where can we trust it?
- 2 A new face solver: the SWMF exact Riemann problem.
- 3 A precision fix, briefly.

The release model

1.5D field-aligned single fluid

$$U = (\rho, \rho u, E)$$

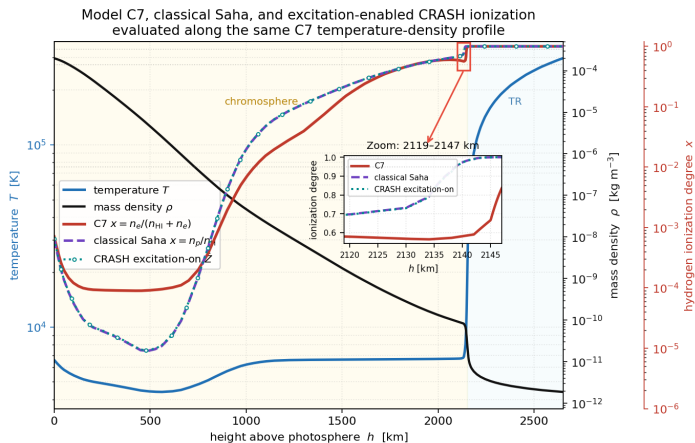
Saha equilibrium EOS

MUSCL-Hancock $\rightarrow$ Godunov

$\rightarrow$ implicit conduction

1600–2153 km, 661 cells

What sets hydrogen ionization in the chromosphere?



- Above ~ 1.8 Mm, Saha **tracks** the semi-empirical C7. Both cross $x = 0.5$ within 7 km.
- Lower down they part — and the **sign flips**: C7 is $10\times$ higher at 500 km, $3\times$ lower at 1200 km.
- Excitation physics does **not** close the gap.

C7 is non-LTE (Avrett & Loeser 2008); the other curves are LTE equilibrium on C7's own (T, n_{H}) .

Two consequences

We adopt Saha equilibrium

- One consistent, differentiable closure: x , n_e , p , e_{int} from (ρ, T) .
- x is **derived** — never stored, never advanced.
- Ionization energy is carried in e_{int} .

We truncate at 1600 km

- A mid-chromospheric reservoir — **not the photosphere**.
- Where we stop claiming the closure applies. Below is *outside the model*.
- Quasi-static, stratified, $v = 0$.

1600 km is a judgement call, not a derived height.

The release face solver: SWMF exact Riemann

Ported, not reinvented:

- `SWMF/util/CRASH/src/test_godunov.f90 :: get_godunov_flux`
- `SWMF/share/Library/src/ModExactRS.f90 :: exact_rs_pu_star, exact_rs_sample`

The Saha internal energy **splits exactly**:

$$e_{\text{int}} = \underbrace{\frac{p}{5/3 - 1}}_{\text{monatomic translation}} + \underbrace{e_{\text{ion}}}_{\text{rides in the offset } E_0}$$

Split $\rightarrow$ **solve** $\rightarrow$ **sample** $\rightarrow$ **carry** E_0

Split $\rightarrow$ solve $\rightarrow$ sample $\rightarrow$ carry

- 1 **Split.** Translational gas at $\gamma = 5/3$; ionization energy and gravitational potential into an offset $E_0 = (E - \frac{1}{2}\rho u^2 - p/(\gamma - 1))/\rho$.
- 2 **Solve** the local *ideal-gas* Riemann problem exactly.
- 3 **Sample** the self-similar solution at $x/t = 0$.
- 4 **Carry** E_0 with the contact, then rebuild the energy flux.

Why freezing composition is the right limit

Chromospheric hydrogen cannot re-equilibrate on a wave timescale: $\tau_{\text{ion}} \sim 10^5$ s in the mid-chromosphere, still $10\text{--}10^3$ s inside shocks — against a dynamical time of ~ 1 minute.

Carlsson & Stein 2002; Leenaarts+ 2007, 2020

Verification

- Toro's **five standard tests**, to the 6 published digits.
- Degenerate cases *exact*: a stationary contact carries identically zero mass.
- E_0 verified to advect passively — no effect beyond round-off.
- 4000s production run: **941 679 108 faces, all solved exactly, zero fallbacks.**

Release validation now fails on a nonzero fallback count.

The bulk EOS is still Saha equilibrium. Composition is frozen only *across the wave interaction* — this is not a finite-rate ionization model.

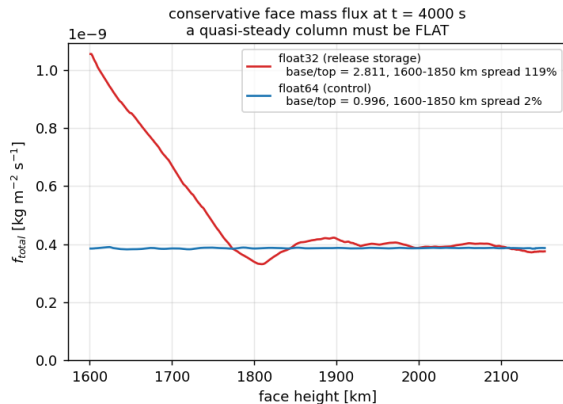
Aside: the lower-chromosphere mass flux was a storage bug

Continuity: a quasi-steady column must carry a **height-independent** mass flux. Ours did not.

$$\rho += \Delta t \cdot \text{rhs}$$

In **619 of 660 cells** that increment is below half a float32 ULP of ρ — it rounds straight back. The density cannot respond.

A double-precision control, everything else identical, makes it flat. Not physics, not the solver, not the boundary — and evaporation at the top moves only $\sim 3\%$.



Status and open questions

Release: single-fluid Saha equilibrium mixture; MUSCL–Hancock + SWMF exact Riemann; implicit conduction; 107 test cases green.

- 1 Saha vs. finite relaxation — the real open question.** We freeze composition *inside* the Riemann problem, then re-impose instantaneous Saha equilibrium *after every step*. Not justified when τ_{ion} is long. Finite-rate network, or a relaxation source?
- 2 Resolution.** N500 long-duration evaporation mass flux is not established as $<10\%$ grid-converged.
- 3 Minor, tracked:** one localized artifact at the first interior face — survives the precision control, so it is a real boundary-closure effect.

Backup: sources

Claim	Where it comes from
C7 vs Saha vs CRASH ionization	<code>util/plot_c7_ionization_profile.py --mode compare</code>
1600 km truncation	<code>docs/doxygen/pages/physics_model.md</code> ; <code>scenarios/model_column.cpp</code>
Relaxation timescales	Carlsson & Stein 2002; Leenaarts et al. 2007; Leenaarts 2020
Toro tests, E_0 passivity, 0 fallbacks	<code>tests/chromo_tests.cpp</code> ; <code>docs/studies/numerics/swmf_godunov_flux_experiment.md</code>
float32 vs float64 control	<code>docs/studies/numerics/float32_precision_control_experiment.md</code>
Figure provenance	<code>figures/README.md</code> ; <code>visualize_commands.md</code> §14–15
Test counts	<code>./build_omp/chromo_tests</code> , current tree

Flux cutover: 022ea34 · Reference-free release: 34ecb00 · Precision control: ac3d9e0